

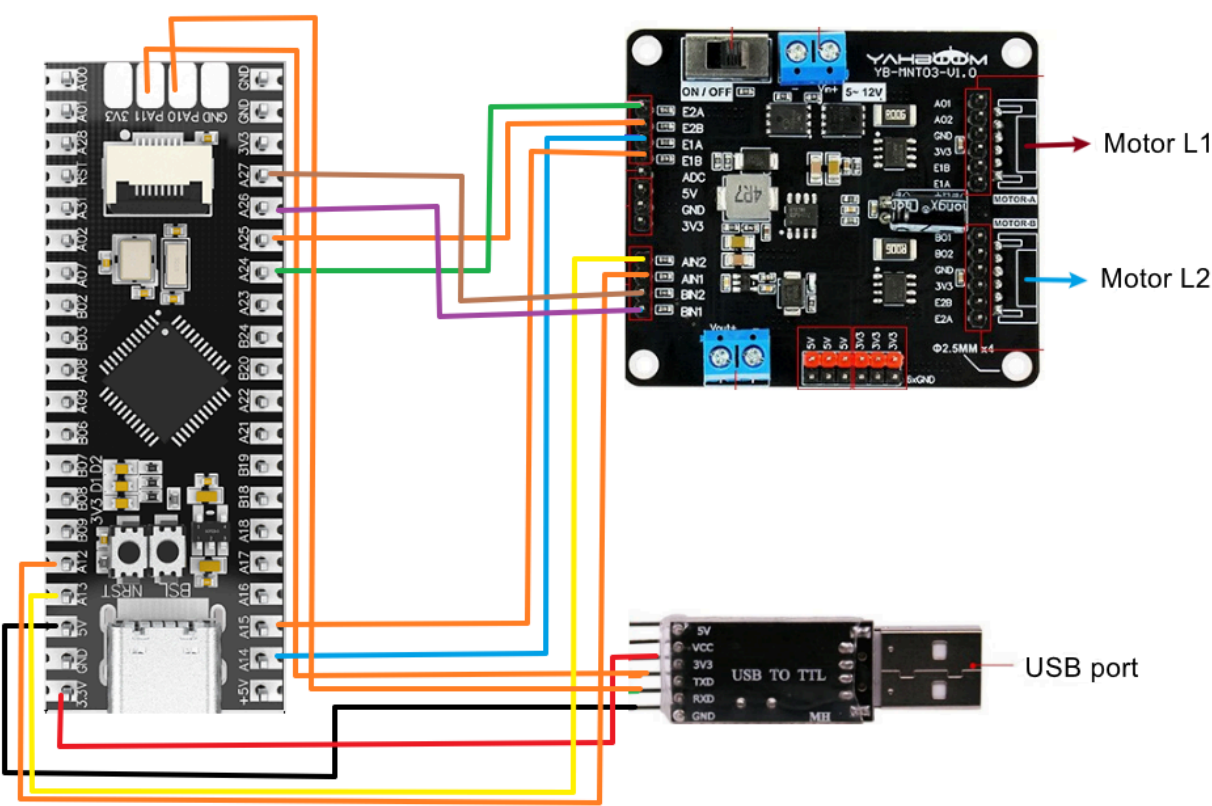
Got encoder data

I. Learning Objectives

Read the motor encoder data.

II. Hardware Connection

MSPM0G3519 and AT8236 pin connection



L1 motor:

MSPM0G3519	AT8236
PA12	AIN1
PA13	AIN2
PA14	E1A
PA15	E1B

L2 motor:

MSPM0G3519	AT8236
PA26	BIN1
PA27	BIN2

MSPM0G3519	AT8236
PA24	E2A
PA25	E2B

Note: You can use the ttl module, or just use the Type-C port of the MSPM0G3519 to connect to the computer.

AT8236 motor driver module pin description :

Pin details					
Interface type	Pin name	Pin description	Interface type	Pin name	Pin description
MCU/ host interface	E1A	Motor 1 Hall signal A	Motor port	AO1	Motor 1 power supply+
	E1B	Motor 1 Hall signal B		AO2	Motor 1 power supply-
	E2A	Motor 2 Hall signal A		GND	GND
	E2B	Motor 2 Hall signal B		3V3	Motor 1 Hall power supply
	ADC	Collect VM input voltage		E1B	Motor 1 Hall signal B
	5V	Output 5V3A power supply		E1A	Motor 1 Hall signal A
	GND	GND		BO1	Motor 2 power supply+
	3V3	Output 3.3V voltage		BO2	Motor 2 power supply-
	AIN1	Motor 1 drive signal 1		GND	GND
	AIN2	Motor 1 drive signal 2		3V3	Motor 2 Hall power supply
	BIN1	Motor 2 drive signal 1		E2B	Motor 2 Hall signal B
	BIN2	Motor 2 drive signal 2		E2A	Motor 2 Hall signal A

III. Program Description

This example sets the baud rate of the serial port printing to 115200 bps.

Software > UART

PROJECT CONFIGURATION...
Project Config... 1/1 ✓ (+)

MSPM0 DRIVER LIBRARY ...

SYSTEM (9)
Board 1/1 ✓ (+)
DMA (+)
GPIO 4 ✓ (+)
MATHACL (+)
Configuration NVM (+)
RTC (+)
SYSCTL 1/1 ✓ (+)
SYSTICK 1/1 ✓ (+)
WWDTC (+)

ANALOG (6)
ADC12 (+)
COMP (+)
DAC12 (+)
GPAMP (+)
OPA (+)
VREF (+)

COMMUNICATIONS (6)
I2C (+)
I2C - SMBUS (+)
MCAN (+)
SPI (+)
UART 1/4 ✓ (+)
UART - LIN (+)

TIMERS (6)

UART (1 of 4 Added) ⓘ

MYUART

Name MYUART
Selected Peripheral UART0

Quick Profiles
UART Profiles Custom

Basic Configuration

UART Initialization Configuration

Clock Source BUSCLK
Clock Divider Divide by 1
Calculated Clock Source 32.00 MHz
Target Baud Rate 115200
Calculated Baud Rate 115211.52
Calculated Error (%) 0.01
Word Length 8 bits
Parity None
Stop Bits One
HW Flow Control Disable HW flow control

- bsp_at8236.h

```
#ifndef __BSP_TB6612_H_
#define __BSP_TB6612_H_

#include "ti_msp_dl_config.h"

void init_motor(void);

void L1_control(uint16_t motor_speed, uint8_t dir);
void L2_control(uint16_t motor_speed, uint8_t dir);
void R1_control(uint16_t motor_speed, uint8_t dir);
void R2_control(uint16_t motor_speed, uint8_t dir);

#endif
```

Define four motor control functions.

- bsp_at8236.c

```
void L1_control(uint16_t motor_speed, uint8_t dir)
```

```

{
    if(dir)
    {
        DL_TimerA_setCaptureCompareValue(PWM_L1_INST, motor_speed, DL_TIMER_CC_0_INDEX);
        DL_TimerA_setCaptureCompareValue(PWM_L1_INST, 0, DL_TIMER_CC_1_INDEX);
    }
    else
    {
        DL_TimerA_setCaptureCompareValue(PWM_L1_INST, 0, DL_TIMER_CC_0_INDEX);
        DL_TimerA_setCaptureCompareValue(PWM_L1_INST, motor_speed, DL_TIMER_CC_1_INDEX);
    }
}
}

```

The L1_control function is used to control the speed and direction of the L1 motor by adjusting the duty cycle of the PWM signal.

motor_speed and dir represent the motor speed and motor direction of the motor respectively.

- bsp_delay.h

```

#ifndef __BSP_DELAY_H_
#define __BSP_DELAY_H_

#include "ti_msp_dl_config.h"

void delay_ms(unsigned int ms);

#endif

```

Define a delay function delay_ms

- bsp_delay.c

```

#include "bsp_delay.h"
#include "ti_msp_dl_config.h"

volatile unsigned int delay_times = 0;

volatile unsigned int getspeed = 10; //Get speed flag, read once every 10ms

extern volatile int32_t gEncoderCount_L1, gEncoderCount_L2;
extern int speed, speed2;

//Accurate ms delay implemented with the SysTick timer
void delay_ms(unsigned int ms) //1ms timing
{
    delay_times = ms;
    while( delay_times != 0 );
}

```

```

//SysTick timer interrupt service function
void SysTick_Handler(void)
{
    if( delay_times != 0 )
    {
        delay_times--;
    }

    if( getspeed != 0 )
    {
        getspeed--;
    }
    else
    {
        getspeed = 10;
        speed = gEncoderCount_L1;
        speed2 = gEncoderCount_L2;
        gEncoderCount_L1 = 0;//clear to 0
        gEncoderCount_L2 = 0;//clear to 0
    }
}

```

It processes the counts of two encoders (L1 and L2) and updates their speed measurements every 10 milliseconds.

- bsp_encoder.c

```

void GROUP1_IRQHandler(void)
{
    //Get interrupt signal
    gpioA = DL_GPIO_getEnabledInterruptStatus(GPIOA,
        L1_Enconder_A_pin_14_PIN | L1_Enconder_B_pin_15_PIN|
        L2_Enconder_A_pin_24_PIN|L2_Enconder_B_pin_25_PIN);

    //If the interrupt is generated by GPIO_EncoderA_PIN_0_PIN
    if((gpioA & L1_Enconder_A_pin_14_PIN) == L1_Enconder_A_pin_14_PIN)
    {
        //Pin14 rising edge, check Pin15 level: low means reverse rotation, high means
forward rotation
        if(!DL_GPIO_readPins(GPIOA,L1_Enconder_B_pin_15_PIN))//P15 is low
        {
            gEncoderCount_L1--;
        }
        else//P15 is high
        {
            gEncoderCount_L1++;
        }
    }

    else if((gpioA & L1_Enconder_B_pin_15_PIN) == L1_Enconder_B_pin_15_PIN)
    {

```

```

        //Pin15 rising edge
        if(!DL_GPIO_readPins(GPIOA,L1_Enconder_A_pin_14_PIN))//P14 is low
        {
            gEncoderCount_L1++;
        }
        else//P14 is high
        {
            gEncoderCount_L1--;
        }
    }
}

```

Used to handle encoder interrupts, read and update encoder data.

- bsp_usart.h

```

#ifndef __BSP_USART_H_
#define __BSP_USART_H_

#include "ti_msp_dl_config.h"
#include "bsp_delay.h"

void uart0_send_char(char ch);
void uart0_send_string(char* str);

#endif

```

A header file for serial communication is defined, which is mainly used to implement simple character and string sending functions.

- bsp_usart.c

```

//Send a single character via serial port
void uart0_send_char(char ch)
{
    //wait while serial port 0 is busy; send the character when it is free
    while( DL_UART_isBusy(MYUART_INST) == true );
    //Send a single character
    DL_UART_Main_transmitData(MYUART_INST, ch);
}

//Send a string via serial port
void uart0_send_string(char* str)
{
    //Current string address is not at the end and the string start address is not null
    while(*str!=0&&str!=0)
    {
        //Send the character at the string start address, and increment the start address
        after sending
        uart0_send_char(*str++);
    }
}

//Serial port interrupt service function

```

```

void MYUART_INST_IRQHandler(void)
{
    //If a serial port interrupt occurs
    switch( DL_UART_getPendingInterrupt(MYUART_INST) )
    {
        case DL_UART_IIDX_RX://If it is a receive interrupt
            //Save the received data into a variable
            uart_data = DL_UART_Main_receiveData(MYUART_INST);
            //Send the saved data back out
            uart0_send_char(uart_data);
            break;

        default://Other serial port interrupts
            break;
    }
}

```

Use USART for serial communication. Includes the functions of sending single characters, sending strings, and a USART interrupt service routine.

- main.c

```

int main(void)
{
    SYSCFG_DL_init();

    //Clear the serial port interrupt flag
    NVIC_ClearPendingIRQ(MYUART_INST_INT_IRQN);
    //Enable the serial port interrupt
    NVIC_EnableIRQ(MYUART_INST_INT_IRQN);
    //
    NVIC_EnableIRQ(GPIO_MULTIPLE_GPIOA_INT_IRQN);//Enable external interrupt
    init_motor();//Start the motor timer

    while (1)
    {
        sprintf(buf, "speed1(10ms):%d\t speed2(10ms):%d\r\n", speed, speed2);
        uart0_send_string(buf);

        L1_control(600,0);//0-1000
        L2_control(400,1);//0-1000

        delay_ms(300);
    }
}

```

Continuously update the speed of the two motors, and set the speed and direction of motors L1 and L2 respectively through the L1_control and L2_control functions. Use the sprintf function to format a message that contains the current speed information of the two motors, and send this message through the uart0_send_string function to monitor or record the status of the motors.

Note: The project source code must be placed in the SDK path for compilation,

For example, the path: D:\TI\M0_SDK\mspm0_sdk_1_30_00_03\1.TB6612

> 新加卷 (D:) > TI > M0_SDK > mspm0_sdk_1_30_00_03				
名称	修改日期	类型	大小	
1.TB6612	2024/7/22 18:59	文件夹		
2.AT8236	2024/7/22 19:47	文件夹		
3.Enconder	2024/7/23 10:36	文件夹		
4.Servo	2024/7/23 11:13	文件夹		
docs	2024/7/23 10:33	文件夹		
examples	2024/7/23 10:34	文件夹		
kernel	2024/7/23 10:37	文件夹		
source	2024/7/23 10:33	文件夹		
tools	2024/7/23 10:33	文件夹		
imports.mak	2024/1/25 11:45	MAK 文件	2 KB	
known_issues_FAQ.html	2024/1/25 11:42	Microsoft Edge ...	67 KB	
license_mspm0_sdk_1_30_00_03.txt	2024/1/25 11:42	文本文档	33 KB	
manifest_mspm0_sdk_1_30_00_03.html	2024/1/25 11:42	Microsoft Edge ...	113 KB	
mspm0sdk_1_30_00_03.log	2024/7/23 10:42	文本文档	5,237 KB	
release_notes_mspm0_sdk_1_30_00_0...	2024/1/25 11:42	Microsoft Edge ...	108 KB	
uninstall.dat	2024/7/23 10:39	DAT 文件	344 KB	
uninstall.exe	2024/7/23 10:39	应用程序	6,048 KB	